Jenil Makwana

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Education

Carnegie Mellon University, School of Computer Science **Aug 2026 – Dec 2027 (expected)**
Master of Software Engineering *Pittsburgh, PA*

- James R. Swartz Entrepreneurial Fellow, 2026–2028
- Coursework: Product Management, Decision Statistics, Formal Methods, Embedded Systems, Computer Systems

University of Wisconsin-Madison **Sep 2021 – Dec 2025**
Bachelor of Science in Electrical Engineering & Computer Science | GPA: 3.5/4.0 *Madison, WI*

- Selected Coursework: Algorithms, Database Management Systems, Building User Interfaces, Mathematical Statistics

Skills

Product: Customer discovery, user requirements, product iteration, product launch, cross-functional delivery
Business & Data: Pricing & packaging, survey analysis, adoption metrics, Python, SQL
AI Product: AI product design, LLM integration, document extraction, reliability workflows, human-in-the-loop review
Technical: TypeScript, Next.js, NestJS, Prisma, PostgreSQL, Stripe, C/C++

Experience

Boxmate **May 2024 – Present**
Founder *Madison, WI*

- Defined Boxmate's two-sided marketplace model, registered the LLC and launched to 150+ UW-Madison students, signing 95 hosts and earning $20K+ in lifetime revenue; worked with lawyers on its terms and conditions.
- Hired and managed an India-based development team and led a cross-functional team of about 10 to ship the Next.js storefront, NestJS/Prisma backend and host mobile app with Stripe checkout and automated host reminders.
- Analyzed 230+ survey responses and customer interviews, finding price sensitivity, low freshman awareness and unmet delivery demand, then set $29/$39/$49 tiers where 88% of 95 paid orders chose Gold, the pickup-and-delivery tier.
- Iterated pricing from per-item to box bundles to flat tiers; summer 2026 revenue grew 3.5x YoY to $15.3K.
- Designed Boxmate's customer flow on iPad and directed its translation into the UI/UX; worked with a designer on the visual identity and launched social media and UGC creator campaigns through SideShift for summer 2026.
- Gave technical and business pitches at Transcend UW, Fetch Rewards and gener8tor gALPHA, winning $17.5K.

USEMCO **Mar 2026 – Jul 2026**
Electrical Engineer *Tomah, WI*

- Identified a manual cutsheet bottleneck and delivered a Python, Next.js and Claude API product adopted by all six engineers across about 18 jobs per week, saving 10–15 minutes per job and flagging missing datasheets.
- Partnered with the EPLAN team on X/Y exports that set datasheet order.
- Designed a two-pass extraction and verification workflow with engineer accept/reject review, feeding corrections back into the AI as electrical rules change; drove adoption of an annual Claude subscription at USEMCO.
- Expanded the tool into searchable job history for reusing schematic sections and tracking cutsheet completion.
- Scoped control-panel orders with sales, resolved client change requests through submittal and resubmittal, and cleared compliance and layout review before releasing EPLAN schematics to the shop floor for assembly.

IdeaFund Ventures **Feb 2026 – Mar 2026**
Software Consultant *Madison, WI*

- Automated deal-flow intake so website pitch submissions populate Monday.com and notify a partner.
- Supported the firm's adoption of Claude for daily workflows.

Credo Product Development **Jun 2024 – Dec 2025**
Software Engineer Intern *McFarland, WI*

- Implemented the hardware watchdog and wrote and executed I2C test procedures for an STM32 leak-detection product shipped to a customer, covering sensor bring-up, fault recovery and bench validation in C.
- Built an internal ESP32 irrigation prototype with moisture sensing, valve control, BLE and a mobile app.
- Wrote bare-metal C firmware for an ATtiny3217 pipe-bender motor controller with a DRV8701 H-bridge, 50 kHz PWM, an RTC-backed 1 ms timer engine with interrupt and polled lists, and 50 ms button debouncing.

UW Space Science & Engineering Center **May 2023 – Jun 2024**
Software Engineer Intern *Madison, WI*

- Replaced manual resistance testing with a configurable Python data pipeline driving a Keithley DAQ6510 and 7700 multiplexer over SCPI/TCP, converting thermistor readings via Steinhart-Hart and logging to HDF5.
- Designed and SPICE-validated a precision heater-amplifier PCB in Altium for the SHIS instrument and prototyped a Zynq FPGA sensor interface whose bare-metal C++ driver halved prototype harness wiring.

Technical Project

FreeRTOS Multisensor Racing Simulator (GitHub) | *PSoC6, FreeRTOS, Embedded C* **Fall 2025**

- Owned game-logic and LCD display work for a team-built simulator, coordinating queue-driven input and display tasks at 20 FPS; wrote the RA8875 driver with double buffering and semaphore-protected SPI.